MAT 3007 – Optimization

Fundamental LP Thorem and SImplex Method

Lecture 06

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Recap

Linear Combination and Span

The linear combination of a set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$ in $\mathbb{R}^n$ with coefficient $a_1, \cdots, a_m \in \mathbb{R}$ is

$$a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \cdots + a_m\boldsymbol{x}_m.$$

The span of this set of vectors is set of all linear combinations:

$$span(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m) := \{a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \cdots + a_m\boldsymbol{x}_m : a_1, \cdots, a_m \in \mathbb{R}\}.$$

Linear Combination and Linear Dependence

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Linear Dependence

A set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$ in $\mathbb{R}^n$ is linearly dependent if there exist scalars $a_1, \cdots, a_m \in \mathbb{R}$, not all zero, such that $\sum_{i=1}^m a_i\boldsymbol{x}_i = \boldsymbol{0}$. We call them linearly independent is they are not linearly dependent.

Basis and Dimension

A set of vectors $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m\}$ in V is called a **basis** of a vector space V if:

- The vectors are linearly independent, and
- They span V, meaning any vector in V can be expressed as a linear combination of $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$.

The **dimension** of V is the number of vectors in a basis of V.

Basis, Kernel and Range

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Kernel and Range of a Matrix A

Kernel (or Null Space) of $\boldsymbol{A}$: The set of all vectors $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{0}$.

Range (or image) of $\boldsymbol{A}$: The set of all vectors $\boldsymbol{b}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$ for some $\boldsymbol{x}$. It is the span of the columns of $\boldsymbol{A}$.

Rank of a Matrix A

The column-(or row-) **rank** of a matrix $\boldsymbol{A}$ is the number of linearly independent columns (or rows).

- Fact: column-rank $=$ row-rank.

The rank of $\boldsymbol{A}$ is defined to be column-(or row-)-rank of $\boldsymbol{A}$.

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Full Column-Rank, Full Row-Rank, and Full-Rank

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

- $\boldsymbol{A}$ has **full column-rank** if all columns of $\boldsymbol{A}$ are linearly independent, i.e., $\text{rank}(\boldsymbol{A}) = n$.
- $\boldsymbol{A}$ has **full row-rank** if all rows of A are linearly independent, i.e., $\text{rank}(\boldsymbol{A}) = m$.
- $\boldsymbol{A}$ is **full-rank** if it is either full column-rank, full row-rank, or both, i.e., $\text{rank}(\boldsymbol{A}) = \min(m, n)$.

Starting Point: Graphical Solutions to LP

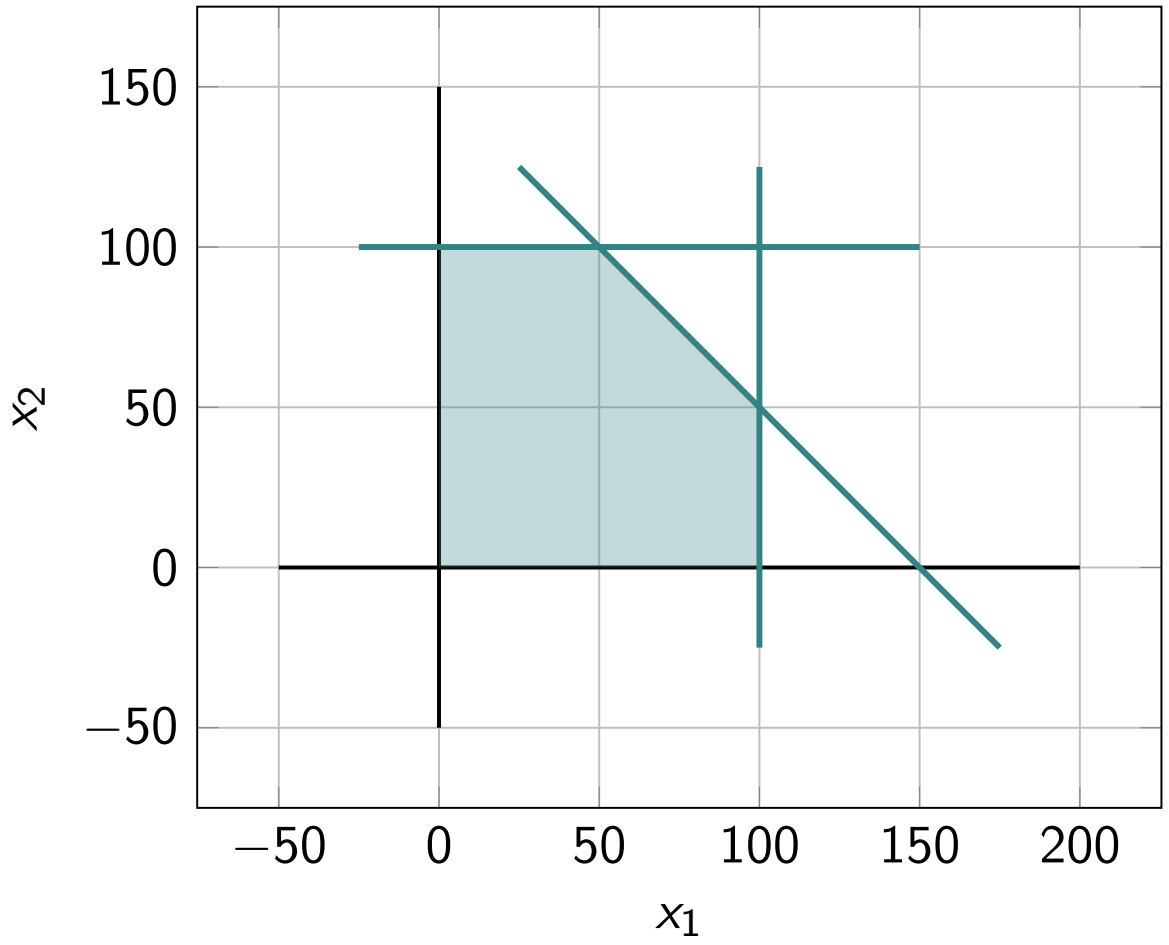
Recall the production problem:

$$
\begin{array}{rrrr}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0.
\end{array}
$$

How can we solve this using a graph?

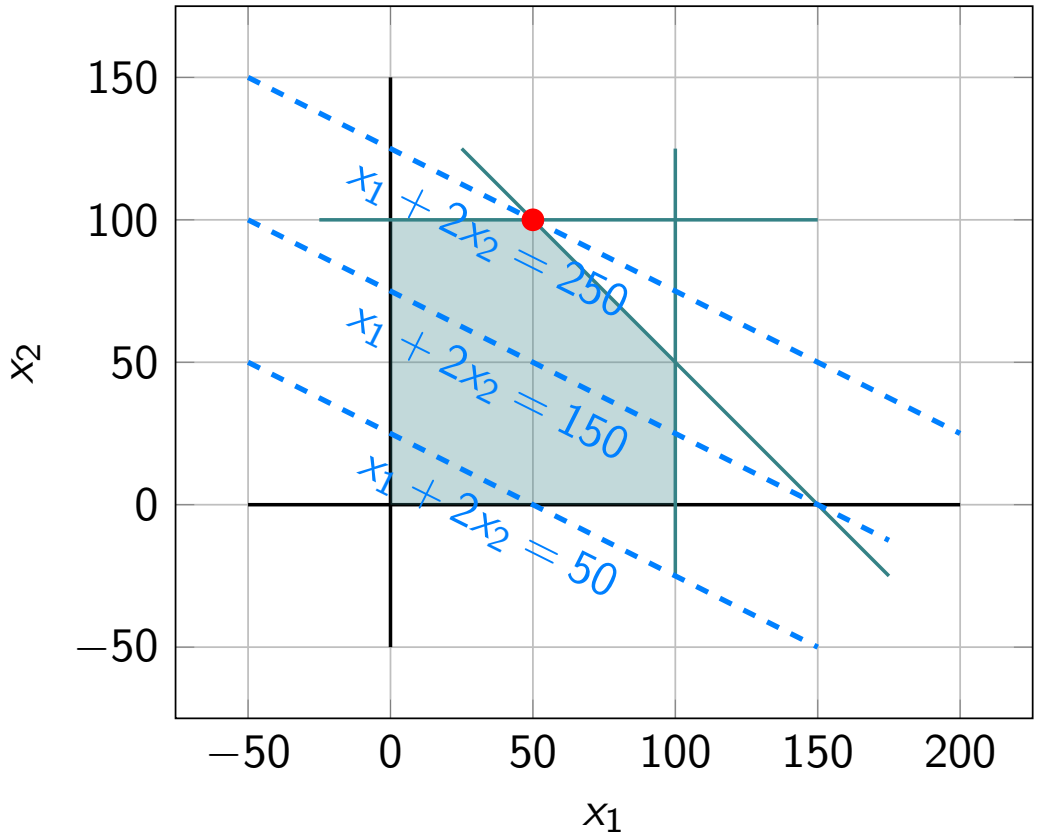
Solve LP from Graph

We first draw the feasible region.



To Maximize $x_1 + 2x_2$ ...

We then draw the function $x_1 + 2x_2 = c$ for different values of c.



- The optimal solution is the highest one among these lines that touches the feasible region.
- **Optimal solution**: $(50, 100)$. **Objective value**: 250.

Some Observations

Observations:

- The feasible region of an LP is a polyhedron.
- The optimal solution tends to be a corner of the feasible region.
- Some constraints are active at the optimal solution ($x_2 \leq 100$, $x_1 + x_2 \leq 150$), some are not ($x_1 < 100$).

Next:

- How to define polyhedron?
- What do we mean by corner geometrically and algebraically?

Some Definitions: Polyhedron

Polyhedron

A polyhedron is a set that can be written in the form:

$$\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}\},$$

where $\boldsymbol{A}$ is an $m \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^m$.

Some Definitions: Polyhedron

Polyhedron

A polyhedron is a set that can be written in the form:

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where $\boldsymbol{A}$ is an $m \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^m$.

- Recall that in the standard form of LP, the feasible set is

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}.$$

- Is this a polyhedron? Why?
- Yes, we can write it as $\boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}$, $-\boldsymbol{A}\boldsymbol{x} \geq -\boldsymbol{b}$, $\boldsymbol{I} \cdot \boldsymbol{x} \geq \boldsymbol{0}$ where $\boldsymbol{I}$ is the identity matrix.

Extreme Points

What do we mean by "corner"? Let's formalize this notion.

Convex Combination

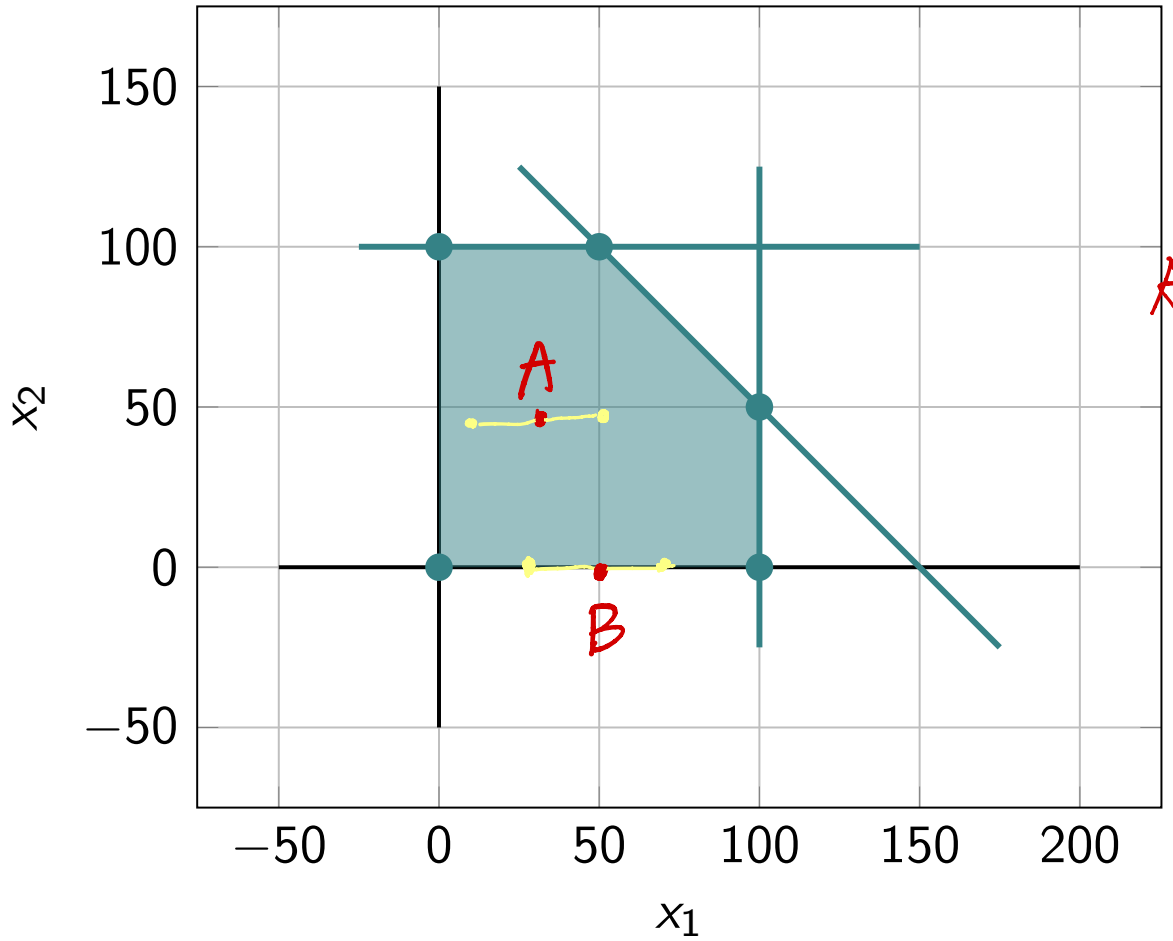
For any $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ and $\lambda_1, \ldots, \lambda_n \geq 0$ satisfying $\lambda_1 + \cdots + \lambda_n = 1$, we call $\sum_{i=1}^{n} \lambda_i \boldsymbol{x}_i$ a convex combination of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$.

Definition: Extreme Point

Let P be a polyhedron. A point $\boldsymbol{x} \in P$ is said to be an extreme point of P if we cannot find two vectors $\boldsymbol{y}, \boldsymbol{z} \in P$ with $\boldsymbol{y}, \boldsymbol{z} \neq \boldsymbol{x}$ and a scalar $\lambda \in [0, 1]$, such that $\boldsymbol{x} = \lambda \boldsymbol{y} + (1 - \lambda)\boldsymbol{z}$.

- That is, $\boldsymbol{x}$ cannot be represented as a convex combination of other two points in P.

Example: Extreme Points



A and B are not extreme points.

How many extreme points are there in this feasible region?

- **Answer:** 5

Finding Extreme Points of an LP

In the following, we consider an LP in its standard form:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{A}$ is an $m \times n$ matrix $(m < n)$ and $\boldsymbol{b} \in \mathbb{R}^m$.

General Assumption:

$\boldsymbol{A}$ has linearly independent rows.

- Recall that rank($\boldsymbol{A}$) = column-rank($\boldsymbol{A}$) = row-rank($\boldsymbol{A}$). Under this general assumption, rank($\boldsymbol{A}$) = m and there are m linearly independent columns.

Extreme Points of LPs: Basic Solutions

Now, we study the extreme points of an LP in their algebraic form.

Definition: Basic Solution

We call $\boldsymbol{x}$ a basic solution of the LP if and only if

1. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
2. There exist indices $B(1), \ldots, B(m)$ such that the columns of

$$\begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

are linearly independent and $x_i = 0$ for $i \neq B(1), \ldots, B(m)$.

Finding a Basic Solution

We write

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}, \quad \boldsymbol{x}_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}.$$

Procedure to Find a Basic Solution:

1. Choose any m independent columns of $\boldsymbol{A}$: $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$.
2. Let $x_i = 0$ for all $i \neq B(1), \ldots, B(m)$.
3. Solve the equation $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ for the remaining $x_{B(1)}, \ldots, x_{B(m)}$. This is equivalent to $\boldsymbol{A}_B \boldsymbol{x}_B = \boldsymbol{b}$ or $\boldsymbol{x}_B = \boldsymbol{A}_B^{-1}\boldsymbol{b}$.

$A = [A_B \; A_N]$ where $N \triangleq \{1, 2, \cdots, n\} \setminus B$

$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix}$

$Ax = [A_B \; A_N] \begin{pmatrix} x_B \\ x_N \end{pmatrix} = A_B x_B + \underbrace{A_N x_N}_{=0} = A_B x_B$

Finding a Basic Solution

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- Since $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$ are linearly independent, $\boldsymbol{A}_B$ is invertible and the last step has a unique solution.
- Basic solution of an LP only depends on its constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

Basic Solutions: Notation

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}, \quad \boldsymbol{x}_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}.$$

- We call $B = \{B(1), \ldots, B(m)\}$ the basic indices for this basic solution, $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$ the basic columns, $\boldsymbol{A}_B$ the basic matrix and $x_{B(1)}, \ldots, x_{B(m)}$ the basic variables.
- We call the remaining indices the non-basic indices, the remaining columns of $\boldsymbol{A}$ the non-basic columns and the remaining variables the non-basic variables.

Basic Feasible Solutions

Definition: Basic Feasible Solution

If a basic solution $\boldsymbol{x}$ also satisfies that $\boldsymbol{x} \geq \mathbf{0}$, then we call it a basic feasible solution (BFS).

How to Find a BFS?

- First find a basic solution $\boldsymbol{x}$.
- Check if $\boldsymbol{x} \geq \mathbf{0}$. $\Longleftrightarrow$ check $x_B \geq 0$

Theorem: Extreme Points and BFS

For the standard LP polyhedron $P := \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \mathbf{0}\}$, the following statements are equivalent:

1. $\boldsymbol{x}$ is an extreme point of P.
2. $\boldsymbol{x}$ is a basic feasible solution.

Proof Sketch

We show how to prove $1 \Rightarrow 2$ (the other part can be found in Chapter 2 of Bertsimas's book). First, if $\mathbf{x}$ is an extreme point, then it must be a feasible solution. In the following, we prove by contradiction that $\mathbf{x}$ must be a basic solution as well.

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Let $B = \{B(1), ..., B(k)\}$ be the set of indices such that $x_i > 0$. If $\mathbf{x}$ is not a basic solution, then it must be that Columns $A_{B(1)}, ..., A_{B(k)}$ are linearly dependent. (Why?)

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Assume they are linearly independent. Since rank of $\boldsymbol{A}$ is m, we have $k \leq m$.

- If $k = m$, by definition of BS, $\boldsymbol{x}$ is a BS.
- If $k < m$, because rank of $\boldsymbol{A}$ is m, we can find another $(m - k)$ indices $S(1), ..., S(m - k)$ such that $A_{B(1)}, ..., A_{B(k)}, A_{S(1)}, ..., A_{S(m-k)}$ are linearly independent. Since $x_i = 0$ for $i \notin \{B \cup \{S(1), ..., S(m - k)\}\}$, $\boldsymbol{x}$ is a BS.

Controdiction.

Proof Sketch

By definition of linear dependence, there must exists $(\alpha_1, ..., \alpha_k) \neq 0$ such that $\alpha_1 A_{B(1)} + \cdots + \alpha_k A_{B(k)} = 0$.

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$$\bar{x}_i = \begin{cases} x_i + \epsilon\alpha_j & i = B(j) \\ 0 & i \notin B \end{cases} \quad \text{and} \quad \tilde{x}_j = \begin{cases} x_i - \epsilon\alpha_j & i = B(j) \\ 0 & i \notin B \end{cases}$$

When ϵ is small enough, $\bar{\boldsymbol{x}}$ and $\tilde{\boldsymbol{x}}$ are feasible. (Why?)

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When ϵ is small enough, $\bar{\boldsymbol{x}}$ and $\tilde{\boldsymbol{x}}$ are feasible. (Why?)

- Since $x_i > 0$ for all $i \in B$, $x_i \pm \epsilon\alpha_j \geq 0$ if ϵ is small enough.
- $\boldsymbol{A}\bar{\boldsymbol{x}} = \boldsymbol{A}_B\bar{\boldsymbol{x}}_B = \boldsymbol{A}_B(\boldsymbol{x}_B + \boldsymbol{\alpha}) = \boldsymbol{A}_B\boldsymbol{x}_B + \boldsymbol{A}_B\boldsymbol{\alpha} = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$, where we denote $\boldsymbol{\alpha} = (\alpha_1 ... \alpha_k)^\top$.

$$A\bar{x} = A_B\bar{x}_B + \underbrace{A_N\bar{x}_N}_{=0}$$

Proof Sketch

By definition of linear dependence, there must exists $(\alpha_1, ..., \alpha_k) \neq 0$ such that $\alpha_1 A_{B(1)} + \cdots + \alpha_k A_{B(k)} = 0$. Construct $\bar{\boldsymbol{x}}$ and $\tilde{\boldsymbol{x}}$ in the following way:

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- $\boldsymbol{A}\bar{\boldsymbol{x}} = \boldsymbol{A}_B\bar{\boldsymbol{x}}_B = \boldsymbol{A}_B(\boldsymbol{x}_B + \boldsymbol{\alpha}) = \boldsymbol{A}_B\boldsymbol{x}_B + \boldsymbol{A}_B\boldsymbol{\alpha} = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$, where we denote $\boldsymbol{\alpha} = (\alpha_1 ... \alpha_k)^\top$.

We can verify $\mathbf{x} = 0.5\bar{\mathbf{x}} + 0.5\tilde{\mathbf{x}}$, which contradicts with that $\mathbf{x}$ is an extreme point. $\square$

Example: Production Planning

Recall the production problem:

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

with standard form:

$$
\begin{array}{rrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & \\
\text{subject to} & x_1 & & +s_1 & & & = 100 \\
& & 2x_2 & & +s_2 & & = 200 \\
& x_1 & +x_2 & & & +s_3 & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0
\end{array}
$$

Example: Continued

We can denote the feasible set by $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix}.$$

- Choose 3 independent columns of $\boldsymbol{A}$, e.g., the first 3 ($B = \{1, 2, 3\}$), we get the corresponding basic solution via:

$$\boldsymbol{x}_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix}.$$

- That is $x_1 = 50, x_2 = 100, s_1 = 50$. Hence, $(50, 100, 50, 0, 0)$ is a basic solution. One can find other basic solutions by choosing other sets of columns.

Example Continued

We can list all basic solutions:

Indices	$\{1,2,3\}$ (A)	$\{1,2,4\}$ (B)	$\{1,2,5\}$ (C)
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Status	feasible	feasible	not feasible

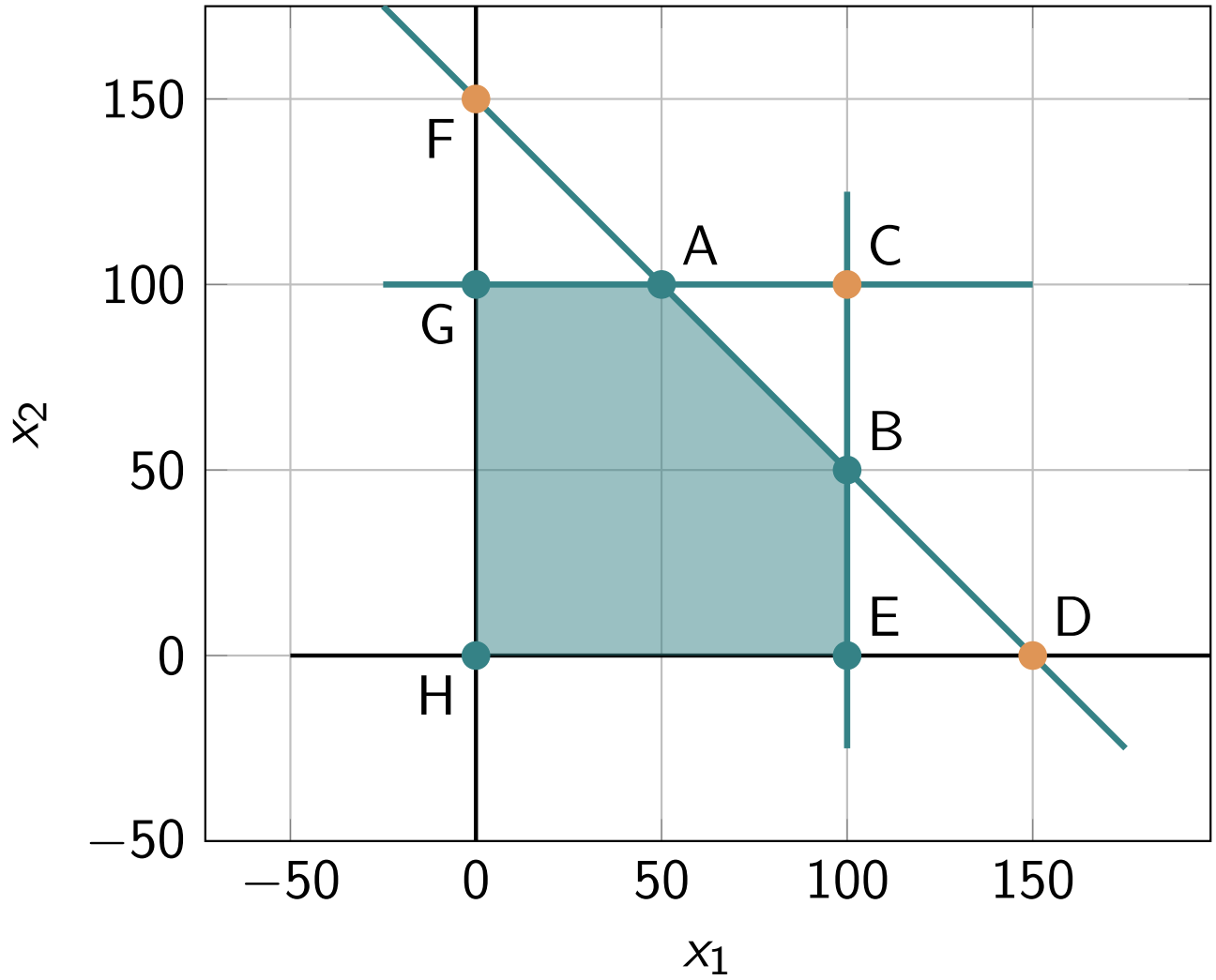
Indices	$\{1,3,4\}$ (D)	$\{1,4,5\}$ (E)	$\{2,3,4\}$ (F)
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Status	not feasible	feasible	not feasible

Indices	$\{2,3,5\}$ (G)	$\{3,4,5\}$ (H)	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Status	feasible	feasible	

The other two choices $\{1,3,5\}$ and $\{2,4,5\}$ lead to dependent basic columns (therefore no basic solutions can be obtained).

Example Continued

The BFSs A, B, E, G, H correspond to all extreme points.



Till now

- Definition of extreme point
- Definition of BS and BFS
- Equivalency between extreme point and BFS in LP

What's missing?

- Why do we care about extreme points and BFSs?

Fundamental LP Theorem

Basic Feasible Solutions

Fundamental LP Theorem

Consider a linear problem in standard form and assume that $\boldsymbol{A}$ has full ~~row~~ rank m.

1. If the feasible set is nonempty, there is a BFS.
2. If there is an optimal solution, there is an optimal solution that is also a BFS.

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Consider a linear problem in standard form and assume that $\boldsymbol{A}$ has full row m.

1. If the feasible set is nonempty, there is a BFS.
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NOT saying all optimal solutions are BFS!

$\rightsquigarrow$ In order to find an optimal solution, we only need to look among BFSs!

Basic Feasible Solutions

Fundamental LP Theorem

Consider a linear problem in standard form and assume that $\boldsymbol{A}$ has full row m.

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Corollary: Characteristics

If an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution with no more than m positive entries. Because a BFS has no more than m positive entries.

Proof Sketch

We first prove part (i). Assume there is a feasible solution $\mathbf{x}$. Let B denote the index set for which $\mathbf{x}$ is positive, and let k be the size of B. In the following, we will show that if $\mathbf{x}$ is not a basic solution, we can always find a solution that has an index set with size less than k. By repeatedly doing this we must be able to find a basic feasible solution.

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why? see the previous proof).

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why? see the previous proof). There exists $\{\{\beta_i\}_{i\in B}\}$ such that $\sum_{i\in B} \beta_i A_i = 0$ and at least one β_i is positive (why?).

if all $\beta_i \le 0$, then $\sum_{i\in B} (-\beta_i) A_i = 0$ and we can use $\{-\beta_i\}_{i\in B}$ as coefficients.

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why? see the previous proof). There exists $\{\{\beta_i\}_{i\in B}\}$ such that $\sum_{i\in B} \beta_i A_i = 0$ and at least one β_i is positive (why?). Define a vector $\boldsymbol{\alpha}$ by

$$\alpha_i = \begin{cases} \beta_i & i \in B \\ 0 & \text{otherwise} \end{cases}.$$

$\epsilon = \min\left\{ \frac{x_i}{\alpha_i} : i \in B \text{ and } \alpha_i > 0 \right\}$

Then we consider $\mathbf{x} - \epsilon\boldsymbol{\alpha}$. There must exist an ϵ such that it is still feasible and that there is at least one more 0 entry in $\mathbf{x} - \epsilon\boldsymbol{\alpha}$. Thus part (i) is proved.

Proof Sketch Continued

Now we consider part (ii). We continue to use the arguments for part (i). Let $\boldsymbol{x}$ be an optimal solution with k positive entries. If x is not a basic solution, then we show that we can always find an optimal solution with fewer than k positive entries (or reach a contradiction). By repeatedly doing this, we must be able to find an optimal basic feasible solution.

We continue to define the same $\boldsymbol{\alpha}$. We claim that it must be that $\mathbf{c}^T\boldsymbol{\alpha} = 0$, otherwise, one can construct a better solution than $\mathbf{x}$ by considering $\mathbf{x} + \epsilon\boldsymbol{\alpha}$ and $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ for some small ϵ, which contradicts with the optimality of $\mathbf{x}$.

(1) if $c^T\alpha > 0$, then $c^T(x-\varepsilon\alpha) = c^Tx - \varepsilon c^T\alpha < c^Tx$

(2) if $c^T\alpha < 0$, then $c^T(x+\varepsilon\alpha) = c^Tx + \varepsilon c^T\alpha < c^Tx$.

If $\mathbf{c}^T\boldsymbol{\alpha} = 0$, then $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ are still optimal solutions. And by previous arguments, we can find one optimal solution with fewer positive entries. □

because x is an optimal solution, and

$c^T(x-\varepsilon\alpha) = c^Tx + \varepsilon c^T\alpha = c^Tx$

From Basic Feasible Solutions to Optimal Solutions



Consequences:

- We only need to search among basic feasible solutions to find the optimal solution.

From Basic Feasible Solutions to Optimal Solutions



Consequences:

- We only need to search among basic feasible solutions to find the optimal solution.

How can we search among the basic feasible solutions?

- One may suggest to list all the BFSs and compare their objective values.

$\rightsquigarrow$ There might be too many BFS!

- For a linear optimization with m constraints and n variables, how many basic feasible solutions can we have?

$\rightsquigarrow$ $C(n, m)$ ... if $n = 1000$, $m = 100$, then this number is 10^{143} ...

The Simplex Method

Simplex Method

The simplex method proceeds from one BFS (a corner point of the feasible region) to a neighboring one to continuously improve the value of the objective function until reaching optimality.

- We need to define the meaning of adjacent or neighboring solutions.
- We need to design an efficient way to find (and move to) the neighboring BFS (e.g., we should try to avoid taking matrix inversions every time).
- We need to design a valid stopping criterion

Neighboring BFS

Definition: Neighboring Basic Solutions

Two basic solutions are neighboring (or adjacent) if they differ by exactly one basic (or non-basic) index.

- For example, a BFS constructed by using the columns $\{1, 2, 3\}$ is a neighbor to the BFS constructed by using the columns $\{1, 3, 5\}$ (but not $\{1, 4, 5\}$).

Example

Consider the production planning problem:

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & & \\
\text{subject to} & x_1 & & +s_1 & & & & = 100 \\
& & 2x_2 & & +s_2 & & & = 200 \\
& x_1 & +x_2 & & & +s_3 & & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0.
\end{array}
$$

The basic (feasible) solutions are given as follows:

Indices	$\{1,2,3\}$	$\{1,2,4\}$	$\{1,2,5\}$
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Objective	-250	-200	infeasible

Indices	$\{1,3,4\}$	$\{1,4,5\}$	$\{2,3,4\}$
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Objective	infeasible	-100	infeasible

Indices	$\{2,3,5\}$	$\{3,4,5\}$	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Objective	-200	0	

Example

Indices	$\{1,2,3\}$	$\{1,2,4\}$	$\{1,2,5\}$
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Objective	-250	-200	infeasible

Indices	$\{1,3,4\}$	$\{1,4,5\}$	$\{2,3,4\}$
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Objective	infeasible	-100	infeasible

Indices	$\{2,3,5\}$	$\{3,4,5\}$	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Objective	-200	0	

If we start from $\{3,4,5\}$ and only allow to move from one BFS to its neighbor (and we move only if the obj. value decreases).

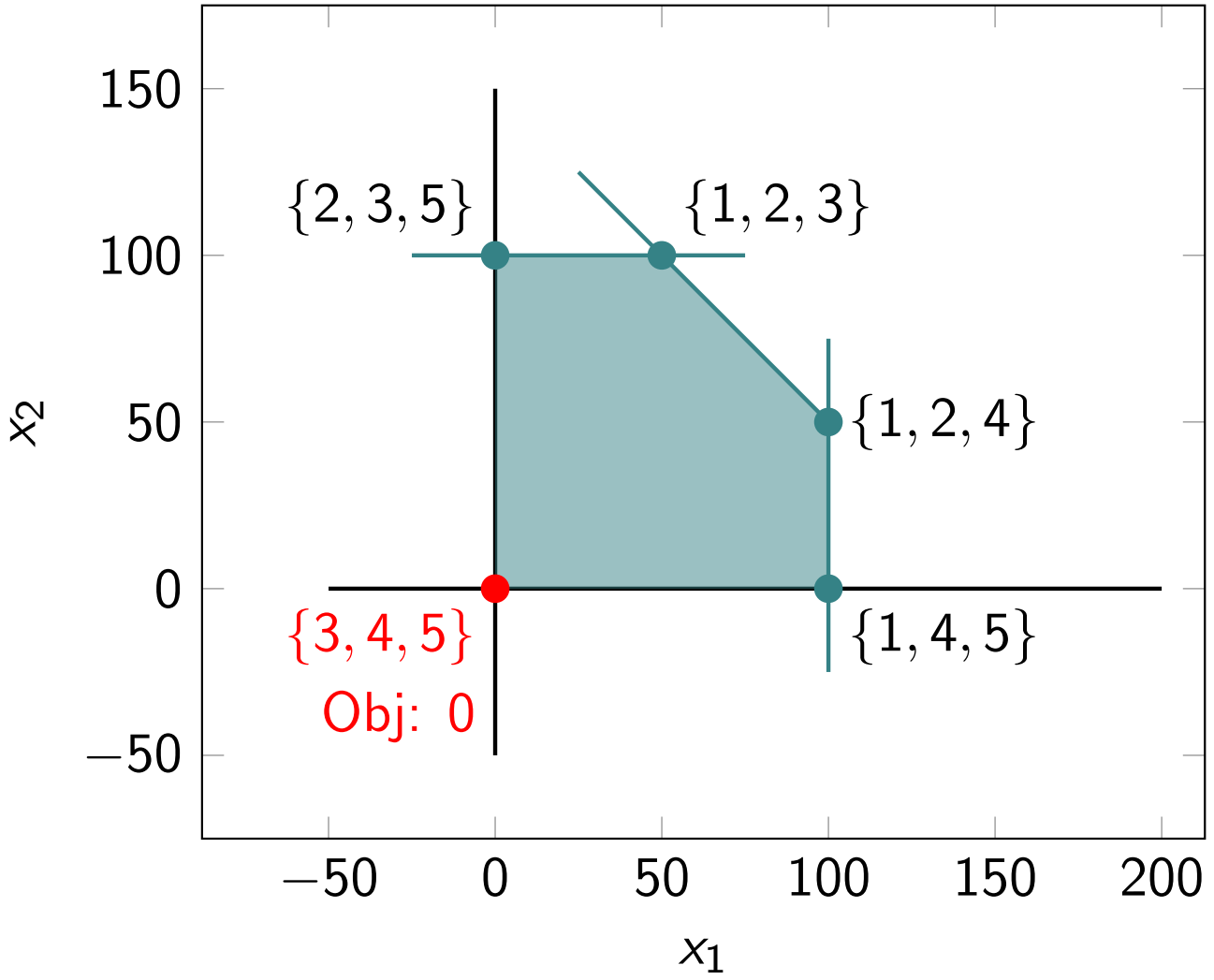
- If we choose $\{1,4,5\}$ first, then we can move along:

$$\{3,4,5\} \to \{1,4,5\} \to \{1,2,4\} \to \{1,2,3\}$$

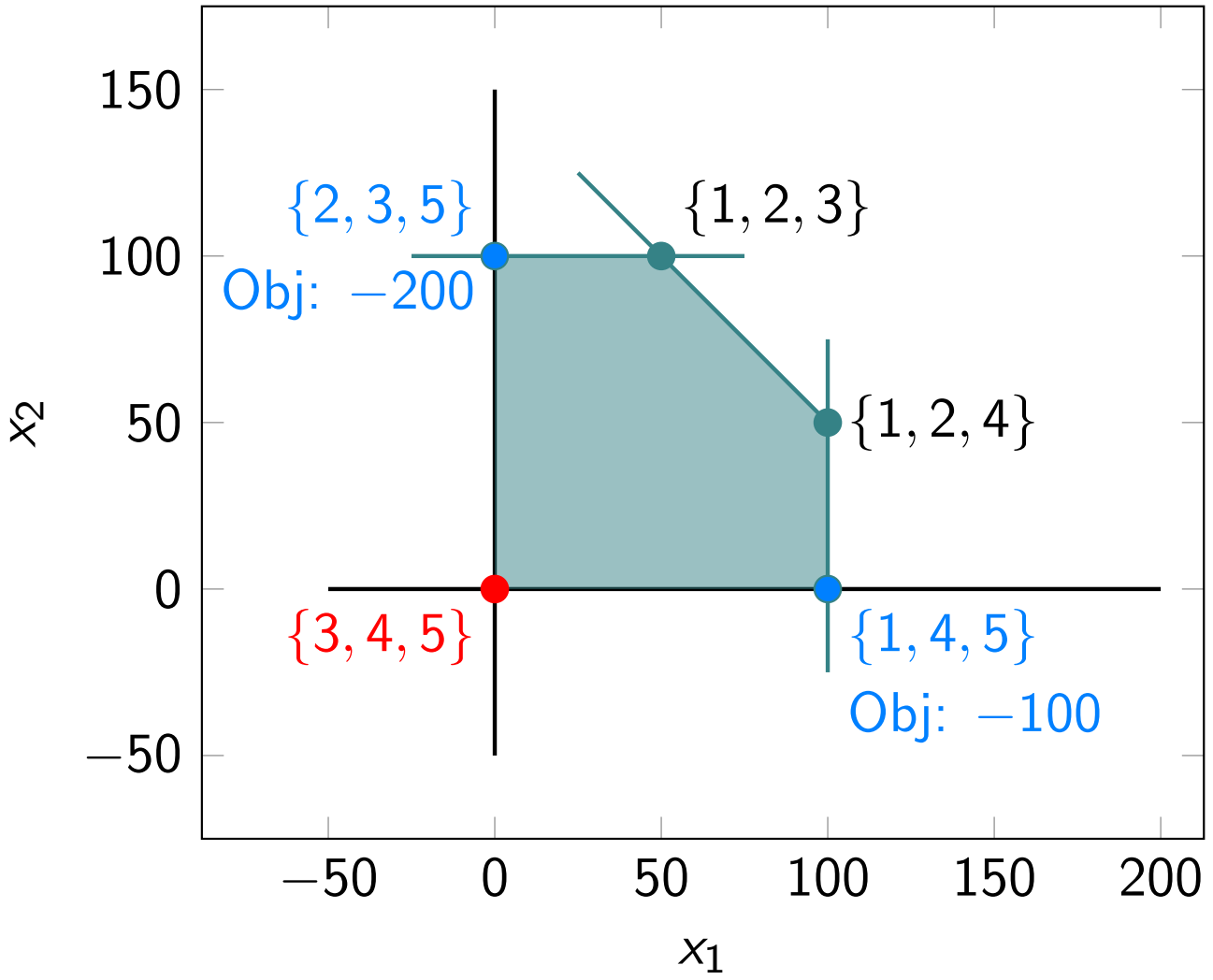
- We can also start from $\{2,3,5\}$, which will give us

$$\{3,4,5\} \to \{2,3,5\} \to \{1,2,3\}$$

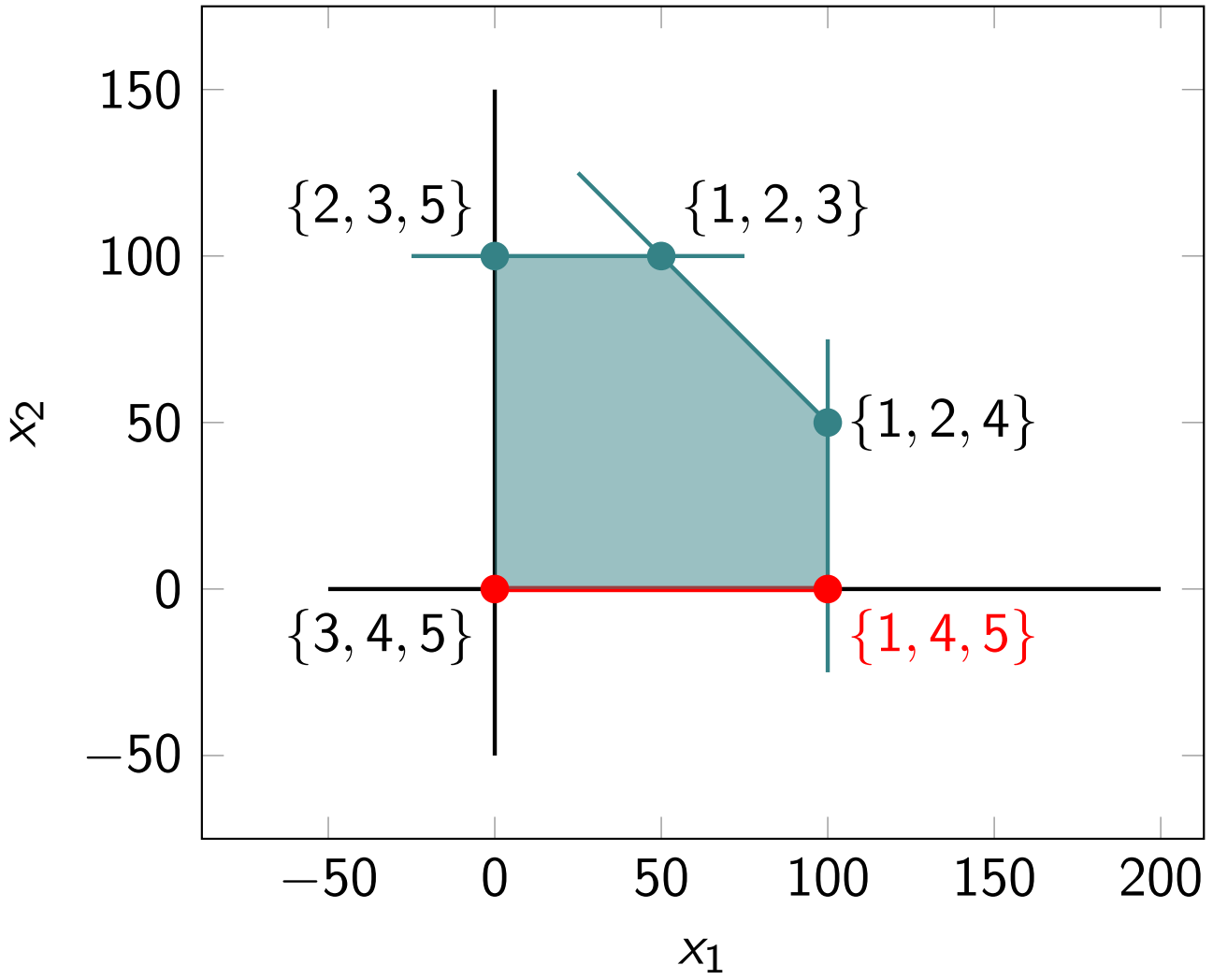
Example



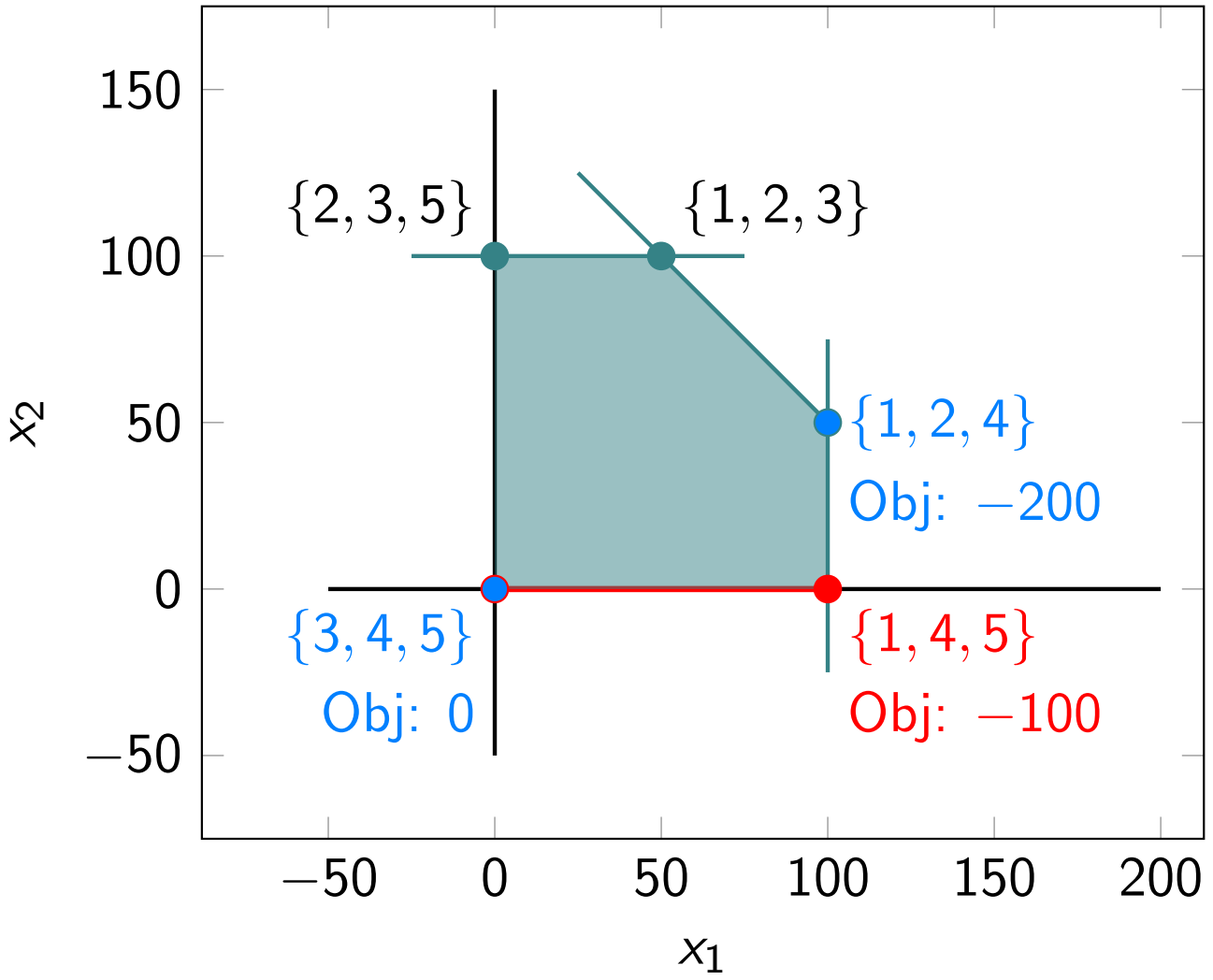
Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
Obj: −200
{1, 2, 3}
{1, 2, 4}
{3, 4, 5}
{1, 4, 5}
Obj: −100

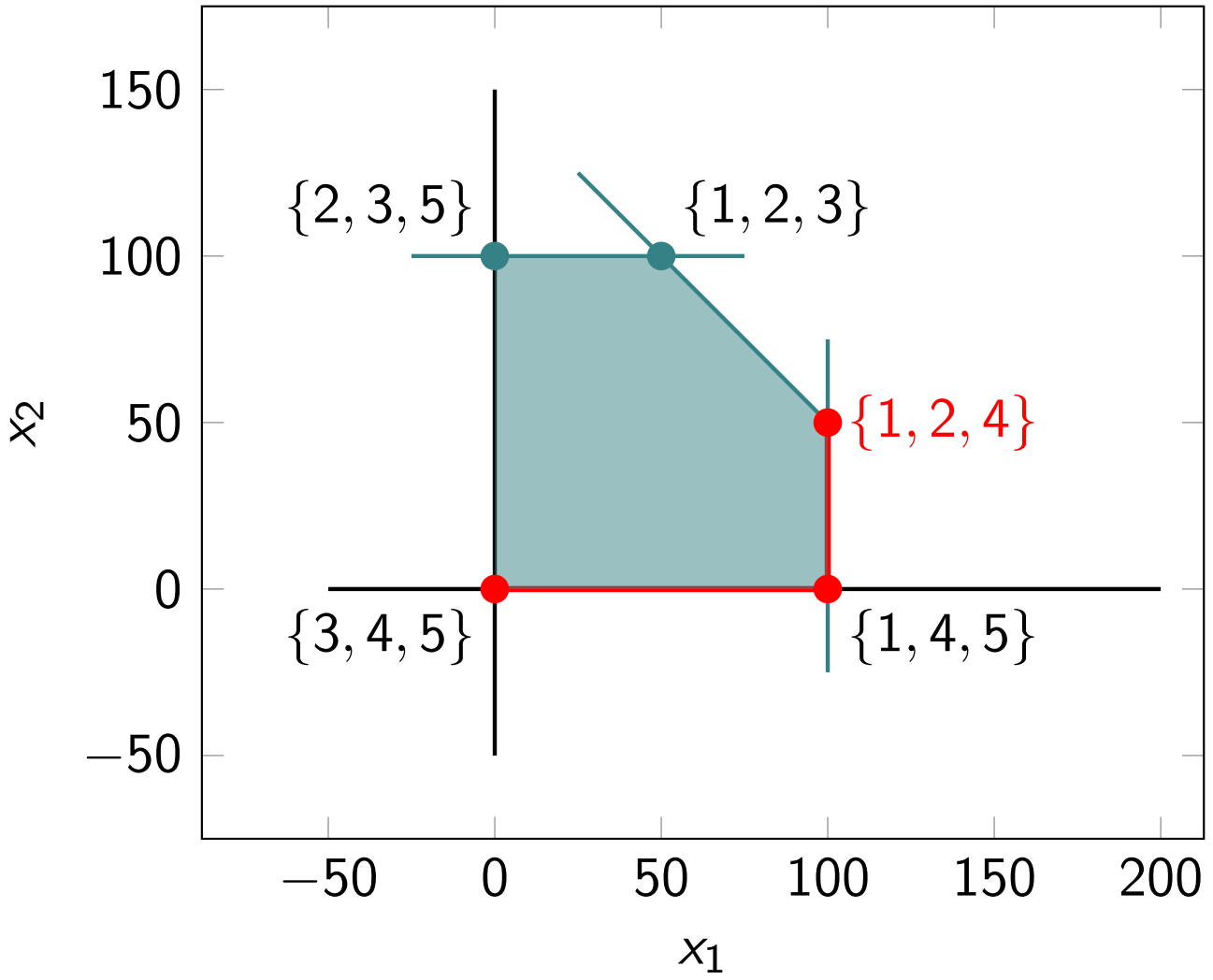
Example



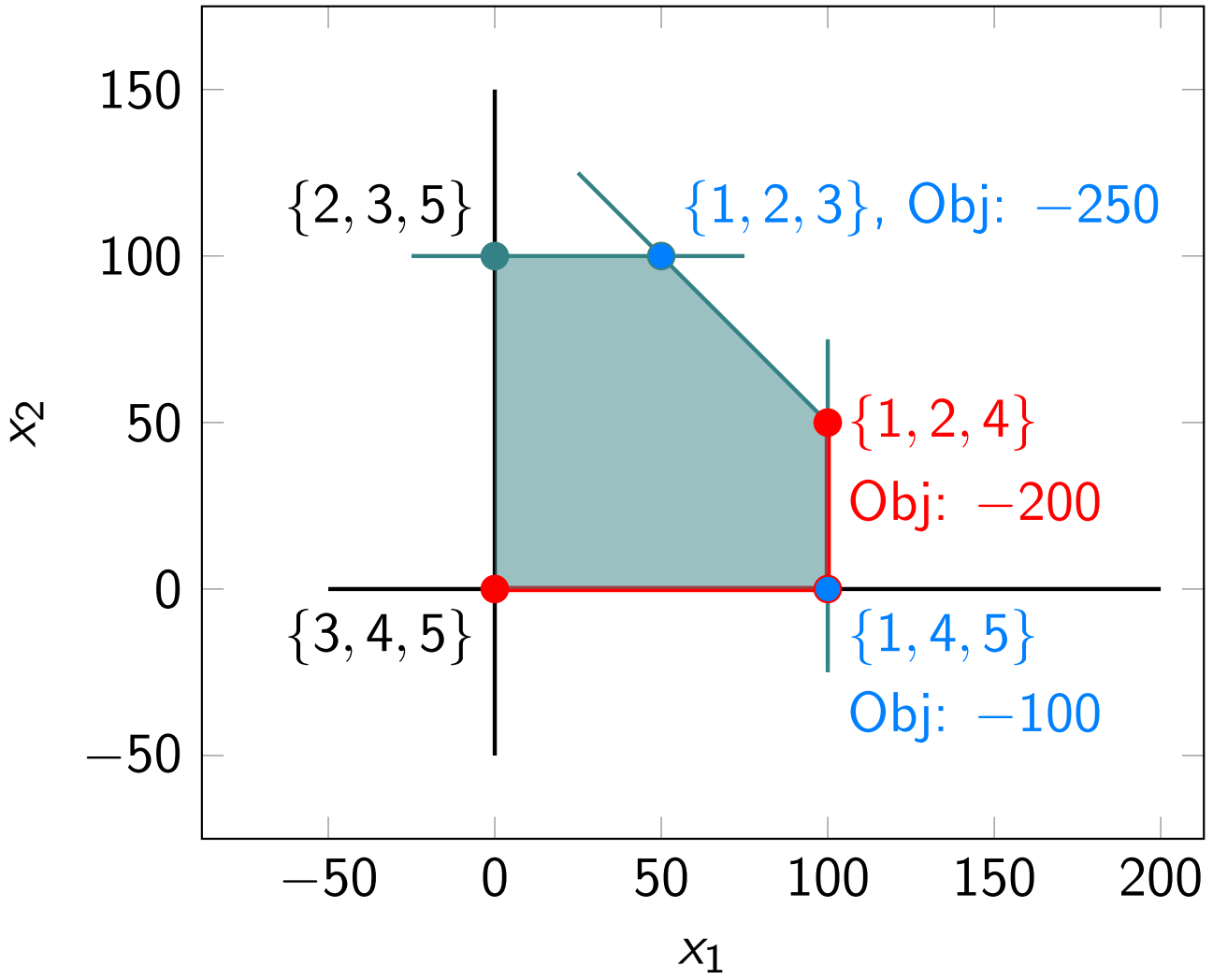
Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
{1, 2, 3}
{1, 2, 4}
Obj: −200
{3, 4, 5}
Obj: 0
{1, 4, 5}
Obj: −100

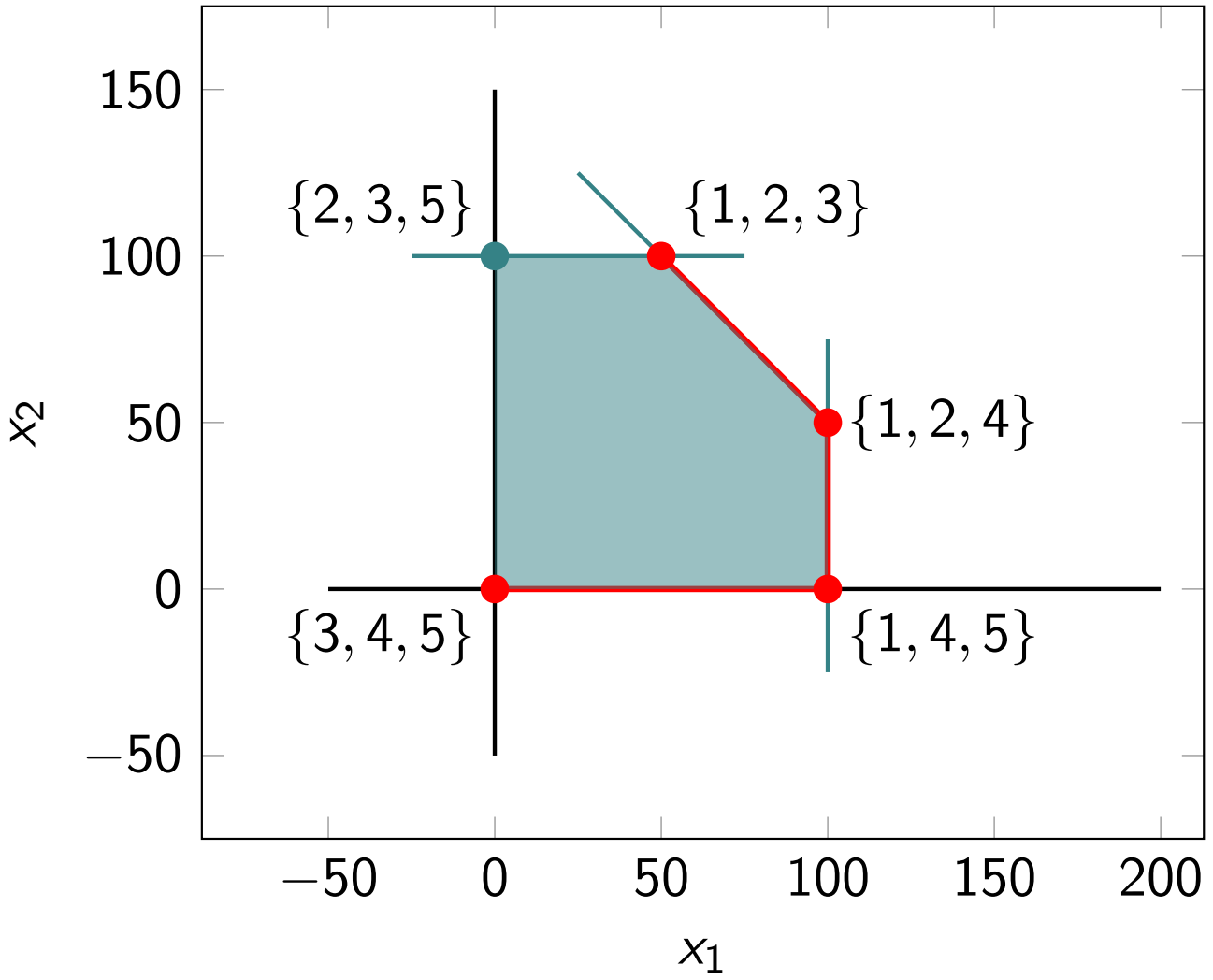
Example



Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
{1, 2, 3}, Obj: −250
{1, 2, 4}
Obj: −200
{3, 4, 5}
{1, 4, 5}
Obj: −100

Example



Checking Neighbors

The main question is how to find a neighboring BFS that improves the current one (reduces the objective function):

- We only need to show how to do it for one step. Then, we can simply iterate the same methods (until we reach optimal).

A naïve way is simply to check all neighbors of the current BFS:

- There are $m(n-m)$ potential neighbors for a BFS (choosing one index to leave, choosing one to enter). For each one, we need to solve a linear equation of m variables.
- This works, but is likely to be very slow.

Finding and Moving to Neighboring BFS

Simplex Method: Starting Point

First, we assume that we have somehow found a BFS whose basis is $B(1), \ldots, B(m)$. Define

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

Denote N the non-basic index set. B and N are disjoint, $|B| = m$, $|N| = n - m$, and $B \cup N = \{1, 2, ..., n\}$. Let $\boldsymbol{A}_N$ be the matrix consisting of the non-basic columns of $\boldsymbol{A}$.

Rearranging the variables, we can write $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, $\boldsymbol{x} = [\boldsymbol{x}_B; \boldsymbol{x}_N]$, where $\boldsymbol{x}_B$ are the basic variables, and $\boldsymbol{x}_N$ are the non-basic variables.

By definition, we have:

$$\boldsymbol{x}_B = \boldsymbol{A}_B^{-1}\boldsymbol{b} \qquad \boldsymbol{x}_N = 0.$$

Finding a Neighboring BFS

We want to find a neighboring BFS:

- Finding a neighbor means changing one of the basic indices.
- We want to select a non-basic variable x_j to enter the basis.
- This means, we want to increase x_j from the current BFS.

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We consider moving $\boldsymbol{x}$ (the current BFS) to $\boldsymbol{x} + \theta \boldsymbol{d}$, $\theta \geq 0$, where:

1. $d_j = 1$.
2. $d_{j'} = 0$ for all other non-basic indices.
3. How to choose $d_{j'} = 0$ for basic indices, i.e., $\boldsymbol{d}_B$?

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2. $d_{j'} = 0$ for all other non-basic indices.
3. How to choose $d_{j'} = 0$ for basic indices, i.e., $\boldsymbol{d}_B$?

What constraints do we have on $\boldsymbol{d}$?

- We need to guarantee that the resulting step $\boldsymbol{x} + \theta \boldsymbol{d}$ is still feasible, that is:

$$\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b} = \boldsymbol{A}\boldsymbol{x} \quad \Longrightarrow \quad \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}.$$

Basic Directions

Now, we write $\boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N]$. Since $d_j = 1$ and $d_{j'} = 0$ for all other non-basic indices, we have

$$\mathbf{0} = \begin{bmatrix} \boldsymbol{A}_B & \boldsymbol{A}_N \end{bmatrix} \begin{pmatrix} \boldsymbol{d}_B \\ \boldsymbol{d}_N \end{pmatrix} = \boldsymbol{A}_B \boldsymbol{d}_B + \boldsymbol{A}_j.$$

$= A_B d_B + A_N d_N = A_B d_B + A_j$

Therefore

$$\boldsymbol{d}_B = -\boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

This means that the direction $\boldsymbol{d}$ is uniquely determined:

$$\boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N] = [-\boldsymbol{A}_B^{-1} \boldsymbol{A}_j; \underbrace{0; \ldots; 1; \ldots; 0}_{d_N}],$$

where the 1 is at the jth entry.

- We call such $\boldsymbol{d}$ the jth basic direction.

Basic Directions

Moving along the jth basic direction currently only guarantees that the equality constraint $\boldsymbol{Ax} = \boldsymbol{b}$ still holds.

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Moving along the jth basic direction currently only guarantees that the equality constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ still holds.

We also need to consider the constraint $\boldsymbol{x} \geq \boldsymbol{0}$. We need to verify that this still holds (i.e., $\boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}$):

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- For non-basic variables i, x_i was 0 and d_i is non-negative (either 1 or 0), therefore, $x_i + \theta d_i$ is still non-negative.
- For basic variables, as long as they were all strictly positive, there must exist a small θ such that $\boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}$.

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- For non-basic variables i, x_i was 0 and d_i is non-negative (either 1 or 0), therefore, $x_i + \theta d_i$ is still non-negative.
- For basic variables, as long as they were all strictly positive, there must exist a small θ such that $\boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}$.

Observation:

- Typically, the basic variables of a BFS are all positive (i.e., m positive entries).
- There are cases that some basic variables in a BFS are equal to 0 (we call it "degeneracy"). We will discuss it later.

Today's lecture

- Fundamental theorem of LP
- The basic idea of Simplex method
- Basic directions

Next lecture

- How to choose stepsize θ to find neighboring BFS?
- How to guarantee the objective value is reducing?
- When to stop?

Questions?